

COLLEGE OF SCIENCE
PHYS2020
Simulation Plan
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0.1 Theory

1 Introduction

When a particle is suspended in a fluid, it undergoes continuous, unpredictable motion as a result of collisions with the surrounding fluid molecules. This phenomenon is known as Brownian motion, named after Robert Brown, who documented it in 1828 [1]. A correct physical explanation was not provided until Delsaux in 1877 [2], who proposed that the motion arose from molecular-scale collisions. A quantitative theoretical framework was produced by Einstein in 1905 [3], who derived that the displacement of a suspended particle along any single axis follows a Gaussian probability distribution. Perrin subsequently verified this experimentally and was awarded the Nobel Prize in Physics in 1926 [4]. Brownian motion provided some of the earliest direct evidence for the atomic nature of matter.

This diffusion is linked to statistical and thermal physics.

Einstein showed that the probability distribution function describing the position of a particle at time t is:

$$f(x) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right), \quad (1)$$

where D , the diffusion coefficient, is given by the Stokes-Einstein relation:

$$D = \frac{k_B T}{6\pi\eta a}. \quad (2)$$

Here T is the absolute temperature, η is the dynamic viscosity of the fluid, and a is the particle radius. For two-dimensional motion observed in the xy plane, the mean squared displacement (MSD) is:

$$\langle r^2 \rangle = \langle x^2 \rangle + \langle y^2 \rangle = 2Dt + 2Dt = 4Dt, \quad (3)$$

where the independence of x and y displacements has been used [5]. Rearranging Eqs. (2) and (3) gives the working equation:

$$k_B = \frac{6\pi\eta a D}{T}. \quad (4)$$

It is also useful to note that Eq. (1) takes the Gaussian form:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad (5)$$

with $\mu = 0$ and $\sigma^2 = 2Dt$. based off of my lab report and the phys BM lab manual

1.1 What I've done so far

I have now set up a github repository which is also linked to my google drive

<https://github.com/carmel3141/Brownian-Motion-Simulation-PHYS2020>

I have tested a few commits to see if i remember how to use github.

yeah so i accidentally bricked my git at one point but its fixed now

moved away from the lattice model to a continuous model, which is more realistic and will be more interesting to analyse after reading a couple of websites

<https://ipython-books.github.io/133-simulating-a-brownian-motion/>

<https://www.quantstart.com/articles/brownian-motion-simulation-with-python/>

one of these has a continuous one where it uses a polar co-ordinate system to generate the random walk which would be interesting to try too

also reference the planning document from the first checkpoint

2 What im currently working on

currently reading :

<https://matplotlib.org/stable/users/explain/animations/animations.html>

to hopefully make this a live simulation, will also plug in some actual physics constants and base the walk off of my theory, and introduce multiple particles.

particle trajectory is now animated but i don't want the animation to trace the path, i want it to draw it, and maybe calculate it live instead? would like the particle simulator in there soon. simulator since adding the animation has doubled up legend for some reason and need to fix. will watch a couple of youtube videos shortly

looking at these:

<https://www.youtube.com/watch?v=CYlsadpjyiY>

<https://www.youtube.com/watch?v=J0fYheMaB-Y>

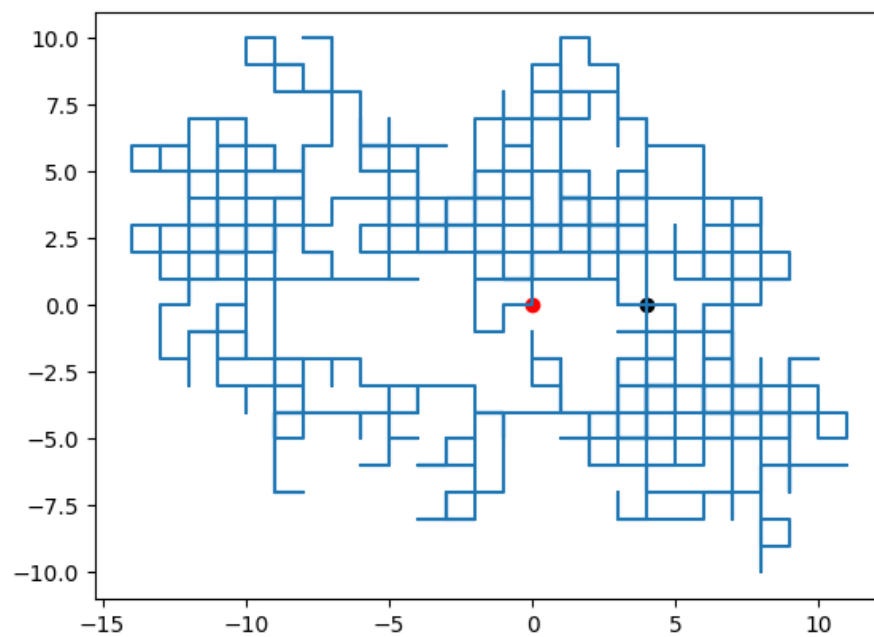
<https://github.com/mominalix/Particle-Physics-Simulator> - something like this (but bm instead of gravity) (using pygame) might work and allow me to extend the scope of the project? now midsems are over i can put more time into the project. i should be able to do the particle simulator with just matplotlib though, would like to do a well with a big particle and some smaller ones to emulate brownian motion, where there is either an input before the simulator begins where you can input (or maybe a slider) different variables such as temperature, viscosity, etc and it will effect the particles and paths. will iterate the function and then compare it to the theoretical values. might be worth investigating what the time complexity is like to see how large we can scale the simulation.

the project was inspired by this:

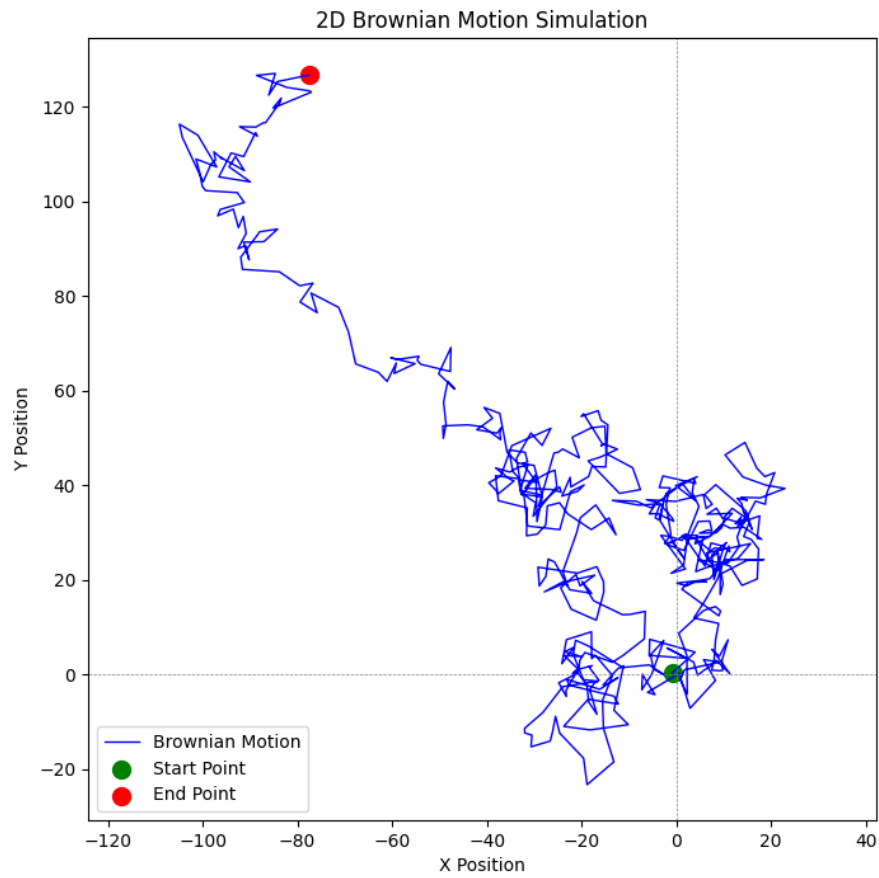
<https://youtu.be/8-B6ryuBkCM?si=9DCAePbyzlZkFZcw>

changed constants to match eta and temperature !! i should go back and revert it to the

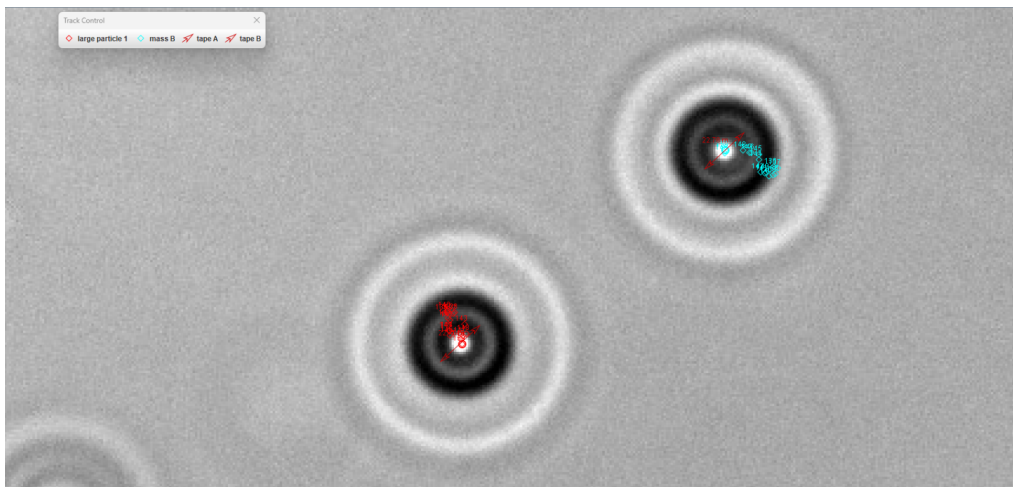
mismatch to see the full numerical difference.



trial 1. added start point and end point, current version has legend too



now uses interpolation instead of grid/lattice so it's more continuous with time-steps, much better !!!



a real photo taken by myself and a lab partner of particles exhibiting brownian motion

References

- [1] R. Brown, “A brief account of microscopical observations on the particles contained in the pollen of plants and the general existence of active molecules in organic and inorganic bodies,” *Edinburgh New Philosophical Journal*, p. 358 (1828).
- [2] J. Delsaux, (1877).
- [3] A. Einstein, “Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen,” *Annalen der Physik (Leipzig)* **17**, 549 (1905). English translation: *Investigations on the Theory of the Brownian Movement*, Dover (1956).
- [4] J. Perrin, “Le mouvement Brownien et la réalité moléculaire,” *Annales de Chimie et de Physique* **18**, 5 (1909). English translation: *Brownian Movement and Molecular Reality*, translated by F. Soddy, Taylor & Francis (1910).
- [5] F. Reif, *Fundamentals of Statistical and Thermal Physics*, McGraw-Hill, New York (1965), Chapter 15.
- [6] Tracker Video Analysis and Modeling Tool, <https://physlets.org/tracker/> (accessed 2026).
- [7] PHYSICS ANU Laboratory Manual, Brownian Motion Australian National University (2026).
- [8] Water viscosity data table, <https://chem-casts.com/data-tables/water-viscosity-data-table> (accessed 2026).
- [9] National Institute of Standards and Technology (NIST), <https://physics.nist.gov/cgi-bin/cuu/Value?k> (accessed 2026).

note: origin of the references are mostly from the lab manual and some are here for future reference